

the signal structure. The general signal structure makes it difficult to characterize mistakes in terms of over or under-updating, unlike in the case of a multivariate normal model. Even when $b = d = 1$ so that automation or own-information bias/neglect are not relevant, an examination of equations (4) and (5) reveals that whether or not a decision-maker exhibits under- or over-updating depends on the difference between $\log \frac{\pi(s^A|\omega=1,s^E)}{\pi(s^A|\omega=0,s^E)}$ and $\log \frac{\pi(s^A|\omega=1)}{\pi(s^A|\omega=0)}$. The propositions above have important implications for the design of human-AI collaboration, which we consider in section 6. Specifically, we study an AI designer who only has access to the AI signal s^A and must decide on one of the three modes of delegation – utilize only the AI prediction, delegate the case to the human, or provide AI assistance to a human expert. The results show that other than in the case when automation neglect is the only relevant bias, the designer must learn the types of biases as well as the distribution of $\pi(s^A, s^E|\omega)$ to determine which delegation modality yields the best decision.

5.3 Estimating Deviations from Bayesian Updating

We now turn to an empirical implementation of the model above. The analysis in this section will be based on designs 2 and 3 because they allow us to observe the same participant make decisions under all information-conditions on a given case. Start by considering an empirical analog to equation (5):

$$\log \frac{p_r(\omega_i = 1|s_{ir}^A, s_{ir}^E)}{p_r(\omega_i = 0|s_{ir}^A, s_{ir}^E)} = a + b \cdot \log \frac{\pi_r(s_{ir}^A|\omega_i = 1, s_{ir}^E)}{\pi_r(s_{ir}^A|\omega_i = 0, s_{ir}^E)} + d \cdot \log \frac{\pi_r(\omega_i = 1|s_{ir}^E)}{\pi_r(\omega_i = 0|s_{ir}^E)} + \varepsilon_{ir}, \quad (7)$$

where we have omitted heterogeneity across radiologists in b_r and d_r . Appendix C.5.6 discusses radiologist heterogeneity in our estimates. Two of the terms in this equation are directly elicited in the experiment: the probability in the second term on the right-hand side, $\pi_r(\omega_i = 1|s_{ir}^E)$, is equal to the radiologists' assessment in the treatment arm where the subjects read cases without AI assistance and the term $p_r(\omega = 1|s_{ir}^A, s_{ir}^E)$ in the dependent variable is the assessment in the treatment arm with AI. The “update term,” given by $\log \frac{\pi_r(s_{ir}^A|\omega_i=1,s_{ir}^E)}{\pi_r(s_{ir}^A|\omega_i=0,s_{ir}^E)}$ will be estimated and substituted into the equation above.

There are three challenges in estimating the update term. The first challenge is that it is a ratio of conditional densities. We address this issue by rewriting it using Bayes' rule as

follows:

$$\log \frac{\pi_r(s_{ir}^A | \omega_i = 1, s_{ir}^E)}{\pi_r(s_{ir}^A | \omega_i = 0, s_{ir}^E)} = \log \frac{\pi_r(\omega_i = 1 | s_{ir}^A, s_{ir}^E)}{\pi_r(\omega_i = 0 | s_{ir}^A, s_{ir}^E)} - \log \frac{\pi_r(\omega_i = 1 | s_{ir}^E)}{\pi_r(\omega_i = 0 | s_{ir}^E)}.$$

If s_{ir}^E can be constructed or controlled for, then we can estimate the first term on the right-hand side using data on ω_i and s_{ir}^A via a binary response model. This estimation is possible because the signal from the AI that the humans receive is isomorphic to the vector of predicted probabilities for the various diseases observed. The second term in this equation has been elicited.

This brings us to the second challenge – constructing s_{ir}^E when estimating $\pi_r(\omega_i = 1 | s_{ir}^A, s_{ir}^E)$ because we do not observe it directly. Let c_i be the patient case associated with case-pathology i and $I(c_i)$ be the set of case-pathologies associated with case c_i . If s_{ir}^E is unidimensional and $\pi_r(\omega_i | s_{ir}^E)$ is monotonic in s_{ir}^E , then $\pi_r(\omega_i | s_{ir}^E)$ is a valid control variable. However, we want to allow for the possibility that the radiologist evaluates a case holistically and uses signals across pathologies. Under the assumption that $s_i^A \perp s_{ir}^E | \omega_i, (\pi_r(\omega_{i'} | s_{i'r}^E))_{i' \in I(c_i)}$, the vector of probabilities for all pathologies reported by r for case i , denoted $(\pi_r(\omega_{i'} | s_{i'r}^E))_{i' \in I(c_i)}$, is a valid control variable. In this multi-dimensional case, a sufficient condition is that s_{ir}^E is invertible in $(\pi_r(\omega_{i'} | s_{i'r}^E))_{i' \in I(c_i)}$. Our empirical specifications will therefore employ multi-variate proxy controls for s_{ir}^E using elicited probability assessments for multiple pathologies. To allow for flexible interactions between s_i^A and s_{ir}^E while avoiding over-fitting, we estimate $\pi_r(\omega_i = 1 | s_{ir}^A, s_{ir}^E)$ using a pathology-specific random forest that predicts ω_i using the vector of predicted probabilities for all pathologies for case c_i reported by radiologist r without AI assistance, the vector of predicted probabilities for case c_i the AI algorithm produces, and summaries of the patient clinical history when made available to the radiologist, and participant-specific fixed-effects. The hyper-parameters of the random forest are chosen by grouped k-fold cross-validation, where we ensure that each patient case appears in only one fold to avoid overfitting to the patient case. Further details of the training procedure are described in appendix C.5.2.

The third challenge is the potential for measurement error, particularly in the form that radiologists' signal s_{ir}^E when elicited without AI might differ from their signal when given AI assistance. Classical measurement error arising from this source would lead to attenuation bias in the coefficient estimates. To address this issue, we will construct instruments s_{ir}^E using the reported probabilities of the other radiologists in our experiment.

With these solutions in hand, we would like to assess whether humans exhibit signal depen-

dence neglect when updating beliefs. As prefaced earlier, although the humans' and AI's signals are not conditionally independent given the ground truth, humans may act as-if they are. We will therefore estimate and select between models that vary the set of signals conditioned on in the update term. For example, in the case when radiologists behave as if s_i^A and s_{ir}^E are independent conditional on ω_i , the update term drops the conditioning on s_{ir}^E . We can vary the pathologies across the set of models considered when constructing $I(c_i)$.²⁹ Including these models in our selection approach will also tell us whether radiologists' assessments are separable across different pathologies.

The correct model of behavior satisfies the conditional moment restriction $E[\varepsilon_{irt} | s_{i,-r}^E, s_i^A] = 0$, where $s_{i,-r}^E$ collects the signals of the radiologists other than r in our experiment. For estimation, we utilize unconditional moment restrictions based on functions of $s_{i,-r}^E$ and s_i^A that closely mimic the terms in equation (7). Our instruments include $\log \frac{\pi(\omega_i=1|s_i^A)}{\pi(\omega_i=0|s_i^A)}$, and leave-one-out averages of $\log \frac{\pi(\omega_i=1|s_i^A, s_{ir'}^E)}{\pi(\omega_i=0|s_i^A, s_{ir'}^E)}$ for radiologists other than r that use various proxies for $s_{ir'}^E$, based on different sets of pathologies $I(c_i)$ that are conditioned on.³⁰ Empirical analogs of the resulting moment conditions are used to estimate the model using GMM.

We will employ the model-selection procedure proposed in [Andrews and Lu \(2001\)](#) to select between non-nested models. This method utilizes the J-statistic of the GMM objective function, which is adjusted for the number of moments and parameters that are included in the model. The selection criterion, MMSC-BIC, penalizes models that reject a greater number of moment restrictions.

5.4 Results

Table 3 presents estimates from six models, which is a subset of the models we consider.³¹ The first three models do not utilize clinical history when made available to the participants to construct the proxy for s_{ir}^E whereas the last three models do. We consider three models in each of these two sets by varying $I(c_i)$. The first corresponds to the case when the signal

²⁹The conditionally independent case corresponds to the extreme case in which $I(c_i) = \emptyset$, whereas the Bayesian model includes all pathologies.

³⁰Specifically, we construct fourteen instruments. The first is a constant and the second is the average of $\log \frac{\pi(\omega_i=1|s_{ir'}^E)}{\pi(\omega_i=0|s_{ir'}^E)}$ for all $r' \neq r$. The remaining twelve construct the average of $\log \frac{\pi(\omega_i=1|s_{ir'}^E)}{\pi(\omega_i=0|s_{ir'}^E)}$ for all $r' \neq r$ by varying the conditioning variables $s_{ir'}$. The different sets of conditioning variables in $s_{ir'}$ are presented in the second panel of appendix table C.21. These sets are used because they are the relevant terms in at least one of the models that we consider in the testing procedure.

³¹See table C.20 in the appendix for the results from all models. Results from all pathologies with AI are qualitatively similar (see table C.21). These analyses were pre-registered except for the model selection exercise, which was not included in the pre-analysis plan.

s_i^A only includes the focal pathology and is conditionally independent of s_{ir}^E given ω_i . The second allows for dependence between s_i^A and s_{ir}^E given ω_i , but only through $\pi(\omega_i | s_{ir}^E)$ for the focal pathology and potentially clinical history. The third utilizes the correct update term constructed using all available AI predictions and the vector of probabilities $\pi(\omega_i | s_{ir}^E)$ for all pathologies. Setting $b = d = 1$ and the constant to 0 in this last case corresponds to Bayesian updating with correct beliefs.

The results from this exercise point to two types of errors in radiologists' use of AI signals. The first type of error is that radiologists neglect signal dependence even though AI predictions and radiologists' signals are highly correlated after conditioning on the ground truth (see appendix table C.18). This conclusion follows because we select the model in column 1, which uses an update term based on equation (6) and therefore does not control for s^E . Another implication of the selected model is that radiologists do not incorporate information across different pathologies since only the focal pathology is relevant. This result validates our previous analysis that analyzes each pathology separately.

Second, across different models, including all models that are not selected, we estimate that radiologists exhibit automation neglect. However, radiologists do not exhibit substantial own-information bias or neglect since the estimated value of d is close to 1 in all specifications. Accounting for measurement error in estimating this model is important, as versions that do not do so estimate d significantly less than 1 (appendix C.5.3).

Connecting these observations back to our theoretical discussion in section 2, the parameters are such that radiologists will not benefit unambiguously from access to the AI signal, primarily because of signal dependence neglect.